

Measuring a neutrino cross section by teaching a computer to re-weight a simulation

A plain-language guide to the unbinned-OmniFold MINERvA analysis

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Abstract

This is a short, jargon-light guide to a measurement of how often muon-neutrinos interact in the MINERvA detector, and to the machine-learning method (“unbinned OmniFold”) used to make it. It explains what the method does in everyday terms, why it lets us measure several quantities at once without an explicitly binned response tensor that grows rapidly with dimension, and what we found — including a faithful reproduction of a published result, a known nuclear-physics effect recovered by a completely independent route, and a study of how far the signal definition can be extended beyond the published muon kinematic cuts. That extension contains both detector-supported and simulation-dominated regions, which are kept separate.

1 What was measured, and why it is hard

A neutrino is a nearly invisible particle: it passes through ordinary matter almost without a trace. To study neutrinos, the MINERvA experiment at Fermilab fired an intense beam of them at a finely segmented detector and recorded the rare occasions when one struck a nucleus and produced visible particles — in this analysis, an interaction that creates a muon (a heavy cousin of the electron). The quantity of interest is the *cross section*: essentially the probability that such an interaction happens, measured as a function of the muon’s motion — how fast it travels along the beam ($p_{\parallel}$) and how much it is deflected sideways (p_T).

The difficulty is that the detector never measures these quantities perfectly. A muon’s recorded momentum is a blurred version of its true momentum, the blur being the detector’s finite resolution and efficiency. The job of *unfolding* is to remove that blur: to recover the true distribution that nature produced from the smeared distribution the detector recorded. It is the same problem as deblurring a photograph, and like deblurring it is delicate — done carelessly it amplifies noise into artifacts.

2 The method, in plain language

The traditional way to unfold builds a big table — a “migration matrix” — that records, for every bin of true momentum, how events leak into neighboring bins of measured momentum, and then tries to invert that leakage. It works, but it has a crippling limitation: each new quantity you want to measure multiplies the number of bins, and the table quickly becomes too large and too sparse to be useful. In practice this caps the method at two — exceptionally three — quantities at a time.

This analysis uses a different, machine-learning method called **OmniFold**. Instead of a table of bins, it trains a computer to compare the real data with a detailed computer simulation, event

by event, and to learn a *weight* for each simulated event — a number that says “there should be a little more (or less) of this kind of event to make the simulation look like the data.” Because the simulation knows both the true and the measured version of every event, that weight can be carried back from the measured world to the true world. Repeating the comparison a few times morphs the simulation until it matches the data, and the morphed *true* simulation is the unfolded answer.

The decisive advantage is what happens when you want to measure more quantities at once. In the table method, adding a quantity multiplies the bins. In OmniFold, adding a quantity is just handing the computer one more column of numbers to learn from — the cost barely grows. That single property is what makes everything in the next section possible.

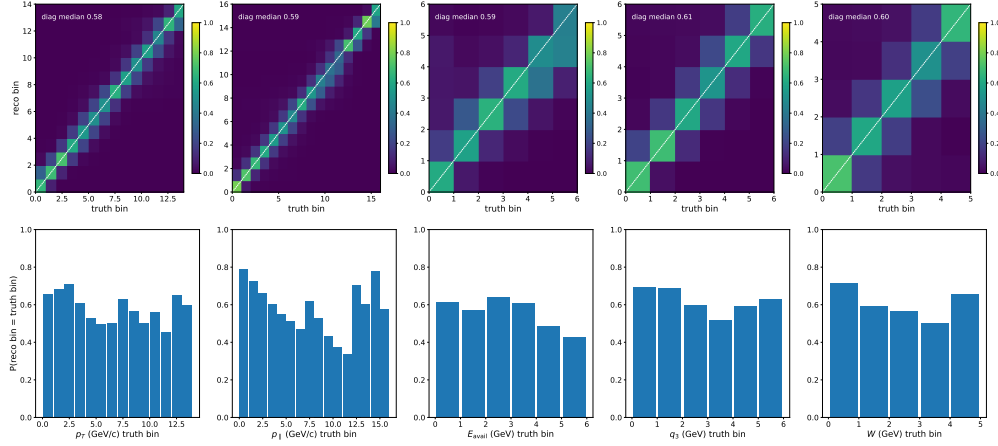


Figure 1: Why unfolding is needed: the detector mixes neighboring values of the true muon momentum together (the off-diagonal spread in these maps). A naïve bin-by-bin correction would be badly biased; unfolding undoes the mixing. OmniFold does it without ever building these bins.

3 What we found

First, a trust check: reproduce a known result

Before trusting a new method, one must show it reproduces an established measurement. MINERvA already published this muon cross section using the traditional method. Our machine-learning result agrees with it closely: the overall interaction rate matches to about one percent (ratio 1.011), and 94.1 % of the individual measurement cells agree within ten percent. Just as important, our *uncertainties* — worked out completely independently — come out essentially the same size as the published ones (6.87 % versus 6.86 % typical). The new method reproduces the old answer and its error bars.

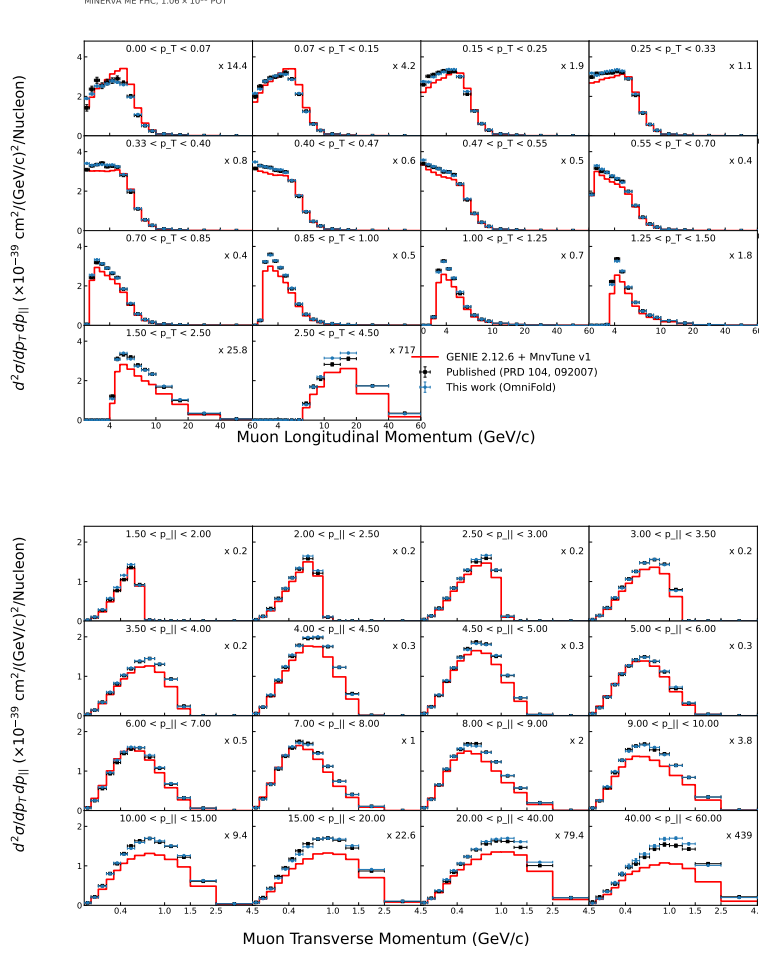


Figure 2: The unfolded muon cross section as a function of the two momentum components. This reproduces a published MINERvA measurement, but was produced by the machine-learning method described above.

Then, measure more at once — and rediscover known physics

With the trust check passed, we used OmniFold’s defining ability and unfolded several quantities *simultaneously* and *without binning* for the first time at MINERvA (earlier MINERvA multi-quantity results used binned methods): the muon momentum together with the available energy (E_{avail} , roughly how much energy the struck nucleus released), the momentum transfer (q_3), and the hadronic mass (W). Each richer measurement, when collapsed back down, reproduces the simpler one — a built-in consistency check.

The first physics payoff appears in the available energy. When we compare the data to four independent state-of-the-art simulations, all four fall short of the data in the low-energy region (Fig. 3). This is not a surprise — it is a known nuclear-physics effect, in which a neutrino occasionally kicks *two* nucleons at once (“2p2h”), that simulations have historically struggled to model. The significance is that our completely independent method rediscovers it: we checked that minor nuclear-physics knobs cannot account for the gap (they move the result by under one percent), whereas adding the missing two-nucleon process fills it in, exactly as expected. And where MINERvA has a dedicated published measurement of this region, ours agrees with it at central-value level on the

maximal common grid. We do not report a covariance-based comparison because the 4D covariance has not been corrected.

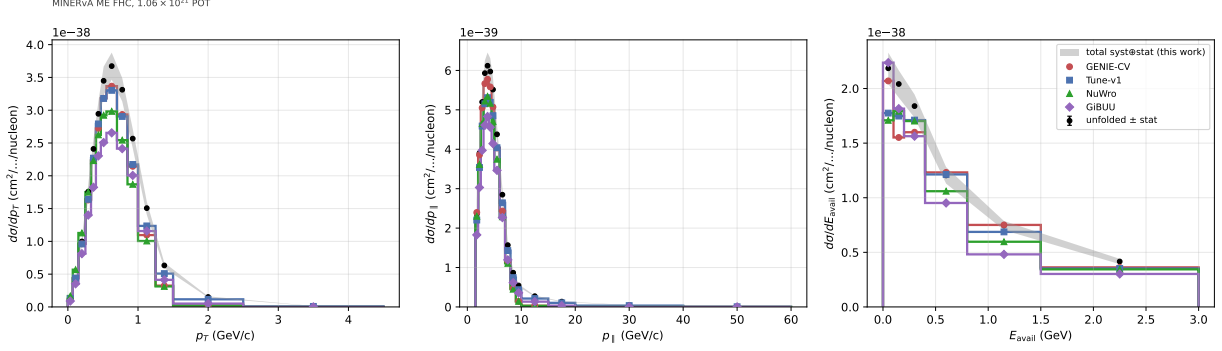


Figure 3: The unfolded central values (black points) compared to four independent simulations. They agree on the muon momentum (left, center) but all fall below the data at low available energy (right) — the known “low-recoil” shortfall, here rediscovered by an independent method. Only the central values are interpreted.

A new puzzle that no simulation explains

Adding the hadronic-mass axis W lets us look where the simulations still miss *after* the two-nucleon process is included. The leftover discrepancy sits in a specific corner: high available energy *and* high W — the “deep-inelastic” region where the neutrino shatters the nucleon. There, all four simulations undershoot the central result. Its statistical significance is not reported without a corrected covariance for this projection. This is the kind of localized discrepancy that points theorists toward what their models are missing — and it was only visible because the unbinned method let us slice the data this finely.

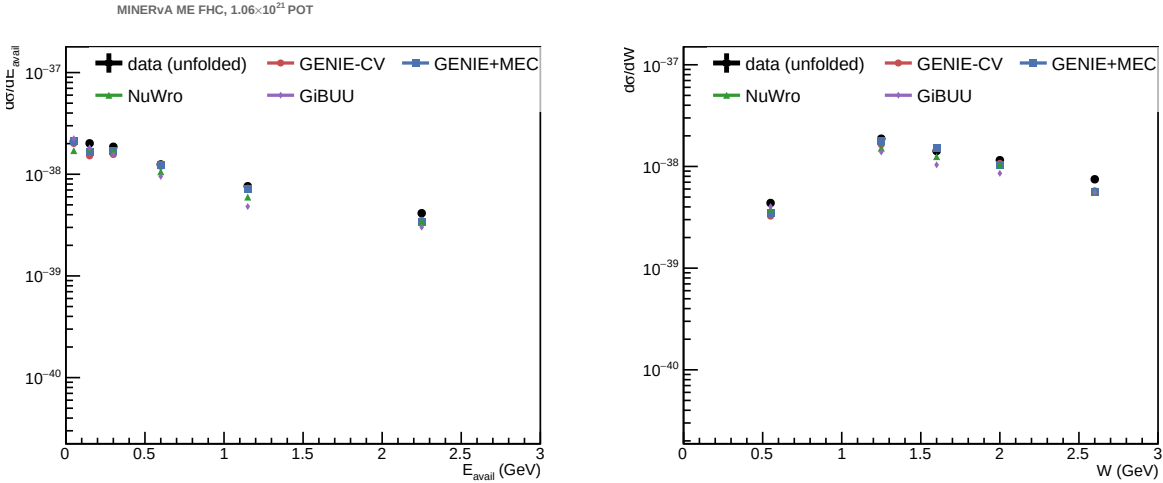


Figure 4: The data versus the simulation band in the available-energy / hadronic-mass plane. Every simulation underpredicts the high-energy, high- W “deep-inelastic” corner; adding the two-nucleon process does not help there.

Alternative representations and signal definitions

First, we ran a “point-cloud transformer” on the non-muon recoil clusters in the detector and the final-state hadrons in the simulation. It provides a useful low-level-input cross-check, but the muon was not given to either classifier; it is therefore a recoil-only cross-check, not an unfold of the joint muon–recoil event. This part of the work is method development rather than a measurement: it shows that the machinery runs end to end on low-level detector data, and it contributes no cross section and no error bar to the results above. A full-event version would supply the muon to the classifiers and test its added information directly. Second, the published signal definition excludes about a third of the simulated interaction rate because of its muon kinematic cuts — a MnvTune-weighted GENIE truth fraction, not a measured one. We relaxed those truth-level cuts while retaining the same tracker interaction volume and reconstructed MINOS-based selection. The preliminary central value is $\sigma_{\text{ext}} = 4.502 \times 10^{-38} \text{ cm}^2/\text{nucleon}$, but only about 6 % of the rate is a newly detector-supported extension; about 28 % lies in low- or zero-efficiency cells and remains a separately labeled, simulation-dependent extrapolation.

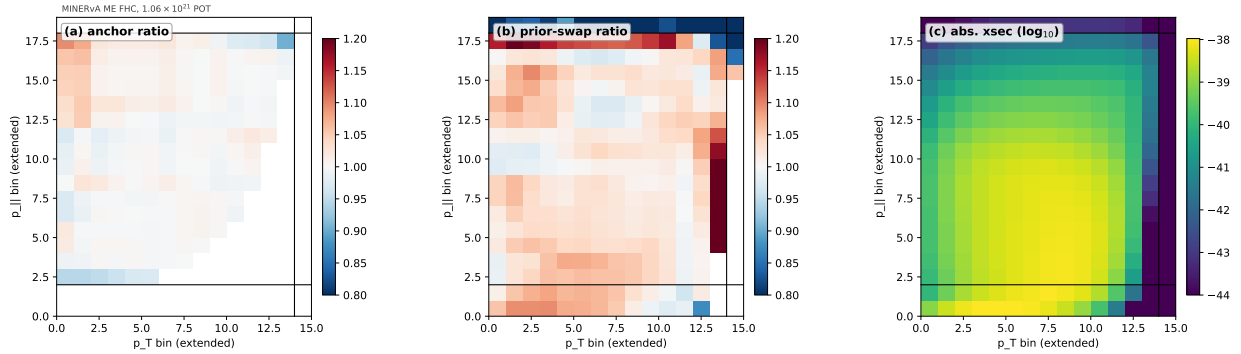


Figure 5: The extended-fiducial scalar study, on the muon momentum grid (p_T versus $p_{||}$, in bin number). Left: inside the published region (between the black lines) the new result matches the original to better than one percent. Center: changing the starting simulation has a small effect in accepted cells but a much larger one in low-efficiency cells. Right: the preliminary central value over the enlarged truth domain (color = rate, on a logarithmic scale); detector-supported and prior-dominated regions must be interpreted separately.

4 The takeaway

Unbinned machine-learning unfolding reproduces a flagship MINERvA measurement *and* does what the traditional method cannot: measure many quantities at once. Using that, this analysis independently confirmed a known nuclear-physics effect, exposed a new discrepancy that all four tested simulations underpredict, and mapped a possible extension of the signal definition while exposing where the detector provides too little information for a model-independent result.